# Lichuan Wu

## PERSONAL DETAILS


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|---|---|
| *Birthday* | July 4, 1986 |
| *Gender* | Male |
| *Address* | Villavgen 16, Department of Earth Sciences, Uppsala University<br>SE-75236 Uppsala, Sweden |
| *Mobile* | +46 (0) 700760015 |
| *E-Mail* | lichuan.wu@geo.uu.se; wulichuan0704@gmail.com |


## EDUCATION

**Ph.D. Meteorology** 2013.07-2017.04
*Uppsala University, Sweden*
Supervisors: Prof. Anna Rutgersson & Erik Sahlée
Thesis Title: Introducing Surface Gravity Waves into Earth System Models

**M.Sc Engineering** 2010.09-2013.06
*Wuhan University of Technology, China*
Supervisor: Prof. Yuanqiao Wen
Thesis Title: Ocean Models Adaptability Comparison and Improvement in the South China Sea

**B.A Engineering** 2006.09-2010.09
*Wuhan University of Technology, China*

## DOCENTSHIP

**Docent (Assoc. Prof.) in Meteorology** 2021.11
*Uppsala University, Sweden*

## PROFESSIONAL EXPERIENCE

**Associate Prof. (Senior lecturer)** 2022.06-
*Uppsala University, Uppsala, Sweden*
**Visiting scholar** 2025.01-2025.06
*University of Reading, UK*
**Assistant Prof. (Associate senior lecturer)** 2019.01-2022-05
*Uppsala University, Uppsala, Sweden*
**Visiting scholar** 2019.01-2019.04
*University of Miami, Miami, USA*
**Researcher** 2017.04-2018.12
*Uppsala University, Uppsala, Sweden*
**Postdoc (80%)** 2017.09-2018.09
*Swedish Meteorological and Hydrological Institute, Norrköping, Sweden*
**Visiting scholar** 2017.08-2017.09
*The Norwegian Meteorological Institute, Bergen, Norway*
Advisor: Prof. Oyvind Breivik
**Visiting scholar** 2016.08-2016.12
*Johns Hopkins University, Baltimore, USA*
Advisor: Dr. Tihomir Hristov
**Teaching Assistant** 2013.07-2017.07
*Uppsala University, Uppsala, Sweden*
**Visiting student** 2011.08-2012.02
*South China Sea Institute of Oceanology, Guangzhou, China*
Advisor: Prof. Shiqiu Peng
**Visiting student** 2010.07 & 2011.08
*Shanghai Typhoon Institute, Shanghai, China*

# TEACHING

- Atmospheric radiation, (5hp, 1ME418, Bachelor/Master course), Course director, Uppsala University, Autumn 2023-
- Remote Sensing with Meteorological Applications, (10hp, 1ME410, Bachelor course), Course director, Uppsala University, Autumn 2019-
- Turbulence and micrometeorology (15hp, 1ME405, Master course), head teacher, Uppsala University, Autumn 2020-
- Atmospheric Physics, (15hp, 1ME401, Master course), head teacher, Uppsala University, Autumn,2019-
- Numerical Modelling of the Atmosphere (1ME406, 10hp, Master course), Uppsala University, Course director, Spring, 2019-
- Project Air Quality Management and Climate (1TV022, 10hp, Bachelor course), Uppsala University, Spring, 2019
- Dynamic Meteorology (NGEA25, 7.5hp, Bachelor course), Lund University, guest lecturer, Autumn, 2017

# SUPERVISION

(a) PhD supervision

1. 2025-now: PhD-Student, Shijie Peng, Uppsala University, "Marginal ice zone", (main supervisor)
2. 2025-now: PhD-Student, Moa Sjberg, Uppsala University, "Marginal ice zone", (main supervisor)
3. 2024-now: PhD-Student, Zhaohui Cheng, Uppsala University, "Wave-ice interactions", (main supervisor)
4. 2021-2025: PhD-Student, Ting Lin, Uppsala University, "Polar lows in the Arctic", (main supervisor)
5. 2021-2025: PhD-Student, Ferran Lopez Marti, Uppsala University, "Atmospheric river", (Co-supervisor)
6. 2020-now: PhD-Student, George Victor Emmanuel, Uppsala University, "wave-ice interactions in marginal ice zones", (main supervisor)
7. 2019-2022: PhD-Student, Wenli Qiao, Zhejiang University, "Momentum flux balance at the air-sea interface", (Co-supervisor)
8. 2018-2021: PhD-Student, Danqin Ren, Uppsala University, "surface wave impact on air-sea interaction", (Co-supervisor)

(b) Postdoc supervision

1. 2024-2026:, Biao Zhao (main supervisor)
2. 2026 - : Zheng Wei (main supervisor)

(c) Visiting PhD student and researchers

1. 2025-2026: Visiting PhD-Student, Yu Gao, Zhejiang University (main advisor)
2. 2022-2023: Visiting PhD-Student, Hongxun Huang, Wuhan University of Technology (main advisor)
3. 2020-2021: Visiting PhD-Student, Xin Peng, Wuhan University of Technology (main advisor)
4. 2018-2019: Visiting PhD-Student, Ting Zhang, Zhejiang University, "Impact of sea surface on wave boundary layer", (main-supervisor)

(c) Master students

1. 2025-2025: Georgios Mastrogiorgos, A numerical case study of the impact of wind farm wakes on ocean surface waves in the Baltic Sea
2. 2022-2024: Di Zhang, Response of polar lows' development to the oceanic temperature profile
3. 2015-2016: Julia Jeworrek, Uppsala University, "Numerical Modelling of Convective Snow Bands in the Baltic Sea Area"

# TRAINING

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| Leading and Developing in Academia | 2025 |
| Supervisory course: Teaching and Assessing Academic Writing | 2022 |
| The Case Method: Teaching with Cases and producing Case material for your teaching | 2020 |
| Student-centred teaching and learning | 2019 |
| Supervising PhD Student | 2018 |
| Writing Research Proposals in English | 2017 |
| Academic Teacher Training Course | 2017 |
| Supervision course: Oral presentation - in theory and practice | 2017 |

## AWARDS

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|---|---|
| Excellent editor of Ocean-Land-Atmosphere Research (OLAR) | 2026.06 |
| Finn Malmgrens stipendiestiftelse | 2021.06 |
| Chinese Government Award for Outstanding Self-financed Student Abroad | 2018.01 |
| Wallenberg Scholarship | 2017.03 |
| Liljewalch Scholarship | (three times)2016,2017,2018 |
| Excellent Master Thesis in Hubei Province | 2014.05 |
| Chinese Government Scholarship for Master's Degree | 2013.05 |
| May 4th Youth medal of Wuhan University of Technology | 2012.04 |
| Merit Postgraduate Student; The First Prize Scholarship | 2012.01 |
| Academic advanced individual of Wuhan University of Technology | 2011.09 |
| Best Paper Award of Asia Navigation Conference 2011 | 2011.03 |
| Excellent graduate of Wuhan University of Technology | 2010.06 |
| The first prize of National Competition of Transport Science and Technology for Students | 2009.04 |

## RESEARCH PROJECTS

1. A stereo video system at stergarnsholm station: exploring the influence of waves and sea surface heterogeneity on air-sea turbulent fluxes, Crafoord grant, (290k sek, PI), 2026.04-2027.04
2. Momentum flux conservation across the air-sea-ice interface within marginal ice zone in Earth System Models (2024-00375), Formas Career Grant, (5.85M SEK, PI), 2025.01-2028.12
3. Turbulence fluxes across the air-sea-ice interface in the marginal ice zone and their influence on polar cyclones (2024-04209). Swedish research council, (4.16M SEK, PI), 2025.01-2028.12
4. Wave attenuation within the marginal ice zone: insight from multi-field campaigns, International sabbatical grant from Uppsala University, (330K SEK, PI), 2025.01-2025.12
5. Wave-Ice Interactions in the marginal ice zone (no. CTS 24:3778), Carl Tryggers Stiftelse, (260k SEK, PI), 2025.01-2027.12
6. Preparation of a training school for uncovering the role of ocean surface waves in the Earth system, Swedish Governmental Agency for Innovation Systems. (357k SEK, PI), 2024.06-2024.12
7. Improving offshore wind energy forecast by exploring the wave-air coupling processes, Carl Tryggers Stiftelse, (60k Euro, PI), 2023.01-2025.12
8. Microwave remote sensing monitoring of summertime sea ice change and the impact from high-latitude synoptic weather systems, (10k Euro, PI), 2023.01-2024.01
9. Greenhouse gas fluxes and earth system feedbacks (GreenFeedBack), Horizon Europe Framework Programme (HORIZON), (7.5M Euro, Co-Applicant), 2022.07-2026.06
10. Polar lows under climate change: a multi-scale coupling chain over the Arctic. Swedish research council, (354k Euro, PI), 2021.01-2024.12
11. Research on Monitoring and Inverting Methods of Air Pollutants Emitted from Ships for the Yangtze River: A Case Study of Wuhan Section, NSFC, (80k Euro, Co-applicant), 2021.09-2024.09
12. Impact of waves on marine atmospheric boundary layer, (6.4k Euro, PI), Research project from Finn Malmgrens stipendiestiftelse, 2021.04-2022.06
13. Ocean wave impacts on the marine atmospheric boundary layer under low wind conditions. Research funding from Qingdao National Laboratory for Marine Science and Technology. (200k rmb, PI), 2019.07-2021.06
14. Air-sea interaction in oceanic extremes. STINT-NSFC, (60k Euro, co-PI), 2019.01-2021.12
15. Air-sea interaction in cold climate. Uppsala University, (50k Euro, PI), 2019.01-2022.12
16. Surface wave impact on air-sea interaction: from micro- to global-scale. FORMAS mobility starting grants, (282.4k Euro, PI), 2018.01-2020.12
17. Impact of waves on offshore wind and wave energy simulation. AFosk funding, (60k Euro, PI), 2017.08-2018.08
18. Introducing surface waves into Earth System models. VR. 2013.07-2017.07
19. Ocean Models Adaptability Comparison and Improvement in the South China Sea. Wuhan University of Technology, (20k RMB, PI), 2011.04-2012.06
20. Study on the Popularization of Navigational Science. Popularization project from China Association for Science and Technology. (30k RMB, PI), 2011.01-2013.01

## PEDAGOGICAL DEVELOPMENT PROJECTS

1. Integration of multiple teaching formats into Project-Based Learning in atmospheric modelling course: an approach toward self-driven learning, Faculty of Science and Technology's University Pedagogical Renewal

Fund, Uppsala University, (220k SEK, PI), 2022.01-2023.12

## JOURNAL PUBLICATIONS

[1] Zhaohui Cheng, Øyvind Breivik, Jean Rabault, Sukun Cheng, Joshua Kousal, Takehiko Nose, Mario Hoppmann, Takuji Waseda, Tsubasa Kodaira, and **Wu, Lichuan***. Wave attenuation by sea ice in the marginal ice zone: Estimation and rollover behavior. *Journal of Physical Oceanography*, 56(11):2507–2521, 2026.

[2] Biao Zhao, Erik Sahle, Jianting Du, and **Wu, Lichuan***. A subgrid-scale parameterization of wave-structure interactions for spectral wave models: Idealized simulations in offshore wind farm conditions. *Journal of Advances in Modeling Earth Systems*, 18(2), 2026.

[3] Josh Kousal, Charles Pelletier, Jasper M.C. Denissen, Luise Schulte, Sarah Keeley, Peter Dueben, Stephen G. Penny, Rae-Seol Park, Diego G. Miralles, Metodija M. Shapkalijevski, Arianna Valmassoi, Ian Renfrew, Lorenzo Zampieri, Rajesh Kumar, Melissa Ruiz-Vsquez, Jan-Peter Schulz, **Lichuan Wu**, and Xabier Pedruzo Bagazgoitia. Numerical weather prediction model coupling strategies, challenges, and outlook. *Bulletin of the American Meteorological Society*, 107(1):E183 – E189, 2026.

[4] Lin Ting and **Wu, Lichuan***. The impact of polar lows on the underlying ocean varies significantly by location. *Communications Earth Environment*, accepted 2025.

[5] **Wu, Lichuan***, Yongqing Cai, and Anna Rutgersson. Ocean surface waves impact on global air-sea co2 flux. *Biogeochemistry*, 168(68), sep 2025.

[6] Yongqing Cai, Ruibo Lei, Dake Chen, Marcel du Plessis, Chengyan Liu, Xianxian Han, and **Wu, Lichuan**. Anticyclonic component of eddy dipoles traps sea ice within the marginal ice zone. *Journal of Geophysical Research: Oceans*, 130(10):e2025JC022426, 2025. e2025JC022426 2025JC022426.

[7] Jian Liu, Keteng Ke, Shimin Yang, Chuang Yang, Zhongyi Sui, Chunhui Zhou, and **Wu, Lichuan**. Evaluation of ship importance in offshore wind farm area based on fusion gravity model in complex network. *Sustainability*, 17(18), 2025.

[8] Chunhui Zhou, Wenjun Jia, Lin Jia, Yuanqiao Wen, Liang Huang, Wuao Tang, Hongxun Huang, and **Lichuan Wu**. Life-cycle assessment of ship electrification in the yangtze river: Environmental and economic trade-offs. *Ocean Engineering*, 341:122831, 2025.

[9] Ferran Lopez-Marti, **Wu, Lichuan**, Gabriele Messori, and Anna Rutgersson. Moisture sources throughout the life cycle of an atmospheric river: Storm dennis case study. *Journal of Geophysical Research: Atmospheres*, 130(16):e2024JD042876, 2025.

[10] Georgios Mastrogiorgos, Johan Arnqvist, and **Wu, Lichuan**. A numerical case study of the impact of wind farm wakes on ocean surface waves in the baltic sea. *Journal of Physics: Conference Series*, 3016(1):012041, may 2025.

[11] Chunhui Zhou, Lin Jia, Wenjun Jia, Wuao Tang, Hongxun Huang, and **Lichuan Wu**. Progress analysis of multi-dimensional perspective research on green ships: A knowledge graph-based approach. *Marine Pollution Bulletin*, 218:118166, 2025.

[12] Yongqing Cai, Ruibo Lei, Dake Chen, and **Wu, L.** Wind-front interaction within oceanic mixed layer following a storm in the southern ocean: Insights from a submesoscale-permitting simulation. *Advances in Atmospheric Sciences*, 2025.

[13] F. Lopez-Marti, M. Ginesta, D. Faranda, A. Rutgersson, P. Yiou, **Wu, L.**, and G. Messori. Future changes in compound explosive cyclones and atmospheric rivers in the north atlantic. *Earth System Dynamics*, 16(1):169–187, 2025.

[14] Lin Ting, , Rutgersson Anna, and **Lichuan Wu***. Influence of mesoscale sea surface temperature anomaly on polar lows. *Environmental Research Letters*, jan 2025.

[15] **Lichuan Wu***, Jinbao Song, Leonie Esters, and Victor Alari. Editorial: Air-sea interaction and oceanic extremes. *Front. Mar. Sci.*, 11, 2024.

[16] Biao Zhao, Erik Sahle, Jianting Du, and **Lichuan Wu***. Wind stress modification by offshore wind turbines: A numerical study of wave blocking impacts. *Ocean Engineering*, 313:119651, 2024.

[17] Chunhui Zhou, Wuao Tang, Zongyang Liu, Hongxun Huang, Liang Huang, Changshi Xiao, and **Lichuan Wu**. Exploring carbon emission reduction in inland port ship based on a multi-scenario model. *Journal of Marine Science and Engineering*, 12(9):1553, 2024.

[18] **Lichuan Wu***, Erik Sahle, Erik Nilsson, and Anna Rutgersson. A review of surface swell waves and their role in airsea interactions. *Ocean Modelling*, 190:102397, 2024.

[19] Biao Zhao, **Lichuan, Wu**, Guansuo Wang, Jun A Zhang, Li Liu, Chang Zhao, Zhanpeng Zhuang, Changshui Xia, Yuhuan Xue, Xinfang Li, et al. A numerical study of tropical cyclone and ocean responses to air-sea momentum flux at high winds. *Journal of Geophysical Research: Oceans*, 129(7):e2024JC020956, 2024.

[20] Daxing Zuo, Chunyi Wu, **Wu, Lichuan**, Yanhui Zheng, Xiaohong Chen, Lina Wang, and Chuanfu Zang. Spatial patterns of short-term extreme precipitation and their causes. *Hydrological Processes*, 38(7):e15227, 2024.

[21] Lin Ting, , Rutgersson Anna, and **Lichuan Wu***. Development of polar lows in future climate scenarios over the barents sea. *Journal of Climate*, 2024.

[22] Lin Ting, Spengler Thomas, Rutgersson Anna, and **Lichuan Wu***. Impacts of sea spray on polar low development. *Quarterly Journal of the Royal Meteorological Society*, 2024.

[23] Mingming Shao and **Lichuan Wu**. Atmospheric fronts shaping the (sub)mesoscale sst-wind coupling over the southern ocean: Observational case. *Journal of Geophysical Research: Atmospheres*, 129(8):e2023JD039386, 2024.

[24] Hongxun Huang, Chunhui Zhou*, Changshi Xiao, Yuanqiao Wen, Weihao Ma, and **Lichuan Wu***. Identification and detection of high nox emitting inland ships using multi-source shore-based monitoring data. *Environmental Research Letters*, 19(4):044051, mar 2024.

[25] Cai Yongqing, Chen Ru, Lei Ruibo, Chen Dake, and **Lichuan Wu**. Factors modulating the forced and intrinsic variability of eddy kinetic energy in the southern ocean from idealized simulations. *Ocean-Land-Atmosphere Research*, 2024.

[26] Shuping Zhang*, **Lichuan, Wu**, Johan Arnqvist, Christoffer Hallgren, and Anna Rutgersson. Mapping coastal upwelling in the baltic sea from 2002 to 2020 using remote sensing data. *International Journal of Applied Earth Observation and Geoinformation*, 114:103061, 2022.

[27] Emmanuel George, Rutgersson Anna, and **Wu, Lichuan***. Role of source terms in parameterizing wave decay in the marginal ice zones. *Ocean Modelling*, 2022.

[28] X. Peng, Y. Wen, **Wu, L.***, L. Huang, C. Zhou, and J. Li. Optimal site selection for remote monitoring sulfur content of ship fuel in ports. *Ocean and Coastal Management*, 2022.

[29] **Wu, L.***, Breivik Oyvind, and Qiao Fangli. The redistribution of air-sea momentum and turbulent kinetic energy fluxes by ocean waves. *Journal of Physical Oceanography*, 2022.

[30] **Wu, L.*** and F. Qiao. Wind profile in the wave boundary layer and its application in a coupled atmosphere-wave model. *Journal of Geophysical Research: Oceans*, 127(2):e2021JC018123, 2022.

[31] M. Gröger, C. Dieterich, J. Haapala, H. T. M. Ho-Hagemann, S. Hagemann, J. Jakacki, W. May, H. E. M. Meier, P. A. Miller, A. Rutgersson, and **Wu, L.** Coupled regional earth system modeling in the baltic sea region. *Earth System Dynamics*, 12(3):939–973, 2021.

[32] Xin Peng, Liang Huang, **Lichuan Wu**, Chunhui Zhou, Yuanqiao Wen, Hualong Chen, and Changshi Xiao. Remote detection sulfur content in fuel oil used by ships in emission control areas: A case study of the yantian model in shenzhen. *Ocean Engineering*, 237:109652, 2021.

[33] Huidong Li*, Björn Claremar, **Wu, Lichuan**, Christoffer Hallgren, Heiner Körnich, Stefan Ivanell, and Erik Sahlée. A sensitivity study of the wrf model in offshore wind modeling over the baltic sea. *Geoscience Frontiers*, page 101229, 2021.

[34] **Wu Lichuan***. Effect of atmosphere-wave-ocean/ice interactions on a polar low simulation over the barents sea. *Atmospheric Research*, 248:105183, 2021.

[35] Wenli Qiao, **Wu, Lichuan***, Jinbao Song, Xue Li, Fangli Qiao, and Anna Rutgersson. Momentum flux balance at the air-sea interface. *Journal of Geophysical Research: Oceans*, 126(2):e2020JC016563, 2021.

[36] Xin Peng, Yuanqiao Wen, **Lichuan Wu***, Changshi Xiao, Chunhui Zhou, and Dong Han. A sampling method for calculating regional ship emission inventories. *Transportation Research Part D: Transport and Environment*, 89:102617, 2020.

[37] T. Olsson, A. Luomaranta, K. Jylhä, J. Jeworrek, T. Perttula, C. Dieterich, **Wu, Lichuan**, A. Rutgersson, and A. Mäkelä. Statistics of sea-effect snowfall along the finnish coastline based on regional climate model data. *Advances in Science and Research*, 17:87–104, 2020.

[38] **Wu, Lichuan***, Mingming Shao, and Erik Sahlee. Impact of Air Wave Sea Coupling on the Simulation of Offshore Wind and Wave Energy Potentials. *Atmosphere*, 11(4):1–21, 2020.

[39] Anna Rutgersson, Heidi Pettersson, Erik Nilsson, Hans Bergstrom, Marcus B.E. Wallin, E. Douglas Nilsson, Erik Sahlee, **Lichuan Wu**, and E. Monica Martensson. Using land-based stations for airsea interaction studies. *Tellus A: Dynamic Meteorology and Oceanography*, 72(1):1–23, 2020.

[40] **Wu, Lichuan***, Øyvind Breivik, and Anna Rutgersson. Ocean-wave-atmosphere interaction processes in a fully coupled modeling system. *Journal of Advances in Modeling Earth Systems*, 11(11):3852–3874, 2019.

[41] **Wu, Lichuan***, Joanna Staneva, Øyvind Breivik, Anna Rutgersson, AJ George Nurser, Emanuela Clementi, and Gurvan Madec. Wave effects on coastal upwelling and water level. *Ocean Modelling*, page 101405, 2019.

[42] **Wu, Lichuan***, Tihomir Hristov, and Anna Rutgersson. Vertical profiles of wave–coherent momentum flux and velocity variances in the marine atmospheric boundary layer. *Journal of Physical Oceanography*, 48(3):625–641, 2018.

[43] **Wu, Lichuan***, Anna Rutgersson, and Erik Nilsson. Atmospheric boundary layer turbulence closure scheme for wind-following swell conditions. *Journal of the Atmospheric Sciences*, 74(7):2363–2382, 2017.

[44] Yongqing Cai, Yuanqiao Wen, **Wu, Lichuan***, Chunhui Zhou, and Fan Zhang. Impact of wave breaking on upper-ocean turbulence. *Journal of Geophysical Research: Oceans*, 122(2):1513–1528, 2017.

[45] Julia Jeworrek, **Wu, Lichuan**, Christian Dieterich, and Anna Rutgersson. Characteristics of convective snow bands along the swedish east coast. *Earth System Dynamics*, 8(1):163, 2017.

[46] **Wu, Lichuan***, David Sproson, Erik Sahlée, and Anna Rutgersson. Surface wave impact when simulating midlatitude storm development. *Journal of Atmospheric and Oceanic Technology*, 34(1):233–248, 2017.

[47] Yuanqiao Wen, Xiaoqiao Geng, **Wu, Lichuan**, Tsz Leung Yip, Liang Huang, and Dingyong Wu. Green routing design in short seas. *International Journal of Shipping and Transport Logistics*, 9(3):371–390, 2017.
[48] **Wu, Lichuan***, Anna Rutgersson, Erik Sahlée, and Xiaoli Guo Larsén. Swell impact on wind stress and atmospheric mixing in a regional coupled atmosphere-wave model. *Journal of Geophysical Research: Oceans*, 121(7):4633–4648, 2016.
[49] **Wu, Lichuan***, Anna Rutgersson, and Erik Sahlée. Upper-ocean mixing due to surface gravity waves. *Journal of Geophysical Research: Oceans*, 120(12):8210–8228, 2015.
[50] **Wu, Lichuan***, Anna Rutgersson, Erik Sahlée, and Xiaoli Guo Larsén. The impact of waves and sea spray on modelling storm track and development. *Tellus A: Dynamic Meteorology and Oceanography*, 67(1):27967, 2015.
[51] Y Cai, Y Wen, and **Wu, Lichuan**. Ship route design for avoiding heavy weather and sea conditions. *TransNav: International Journal on Marine Navigation and Safety of Sea Transportation*, 8, 2014.
[52] **Wu, Lichuan**, Yuanqiao Wen, Chunhui Zhou, Changshi Xiao, and Jinfeng Zhang. Modeling the vulnerability of waterway networks. *Journal of waterway, port, coastal, and ocean engineering*, 140(4):04014012, 2014.
[53] Zhengqiang Li, Yuanqiao Wen, **Wu, Lichuan**, Changshi Xiao, Chunhui Zhou, Xiaochun Wu, and Dingyong Wu. Characteristic analysis of wind zone climate change in the north pacific ocean route. *Journal of Dalian Maritime University*, (2):51–55, 2014. (In Chinese with English Abstract).
[54] **Wu, Lichuan**, Yuanqiao Wen, Shiqiu Peng, Jinfeng Zhang, and Changshi Xiao. Optimizing the routing of ships tropical cyclone avoidance based on the numerical forecasts. *Natural hazards*, 69(1):781–792, 2013.
[55] Xue Yang, Yuanqiao Wen, Liwen Huang, and **Wu, Lichuan**. Relative humiditys climate change and its impacts on navigation of middle yangtze river. *Journal of Wuhan University of Technology(Transportation Science & Engineering)*, 37(3):622–626, 2013. (In Chinese with English Abstract).
[56] **Wu, Lichuan**, Shigang Lv, Shanshan Du, and Yuanqiao Wen. Wind fields climate change and its impacts on navigation of middle yangtze river. *Journal of Wuhan University of Technology(Transportation Science & Engineering)*, 36(02):403–406, 2012. (In Chinese with English Abstract).
[57] Jianfeng Dai, **Wu, Lichuan**, Shijun Hu, Y Ji, and Yuanqiao Wen. Research on the vulnerability of waterway traffic system against typhoon disaster in zhejiang province. *Journal of Wuhan University of Technology*, 34(04):63–66, 2012. (In Chinese with English Abstract).
[58] Shijun Hu, Y Ji, **Wu, Lichuan**, and Yuanqiao Wen. Analysis of the time-space characteristic of the waterway typhoon disaster in zhejiang. *Journal of Wuhan University of Technology(Transportation Science & Engineering)*, 36(04):782–786, 2012. (In Chinese with English Abstract).
[59] **Wu, Lichuan**, Yuanqiao Wen, and Peiyan Chen. Risk analysis of wind field in navigation environment of zhejiang province based on the information diffusion theory. *Journal of Wuhan University of Technology(Transportation Science & Engineering)*, 35(05):1010–1014, 2011. (In Chinese with English Abstract).
[60] **Wu, Lichuan**, Yuanqiao Wen, Peiyan Chen, and Y Ji. Pre-assessment model of tropical cyclone disaster on navigation environment of costal port. *Journal of Wuhan University of Technology*, 33(07):73–80, 2011. (In Chinese with English Abstract).
[61] Fucai Jiang, **Wu, Lichuan**, and Yuanqiao Wen. Wind fields climate change of zhejiang costal area navigation environment. *Journal of Wuhan University of Technology*, 32(07):173–177, 2010. (In Chinese with English Abstract).
[62] Yuanqiao Wen, **Wu, Lichuan**, and Peiyan Chen. Tropical cyclones' impact frequency study on sea routes of china costal area during 1949-2008. *Journal of Dalian Maritime University*, 36(02):27–30, 2010. (In Chinese with English Abstract).
[63] Yuanqiao Wen, **Wu, Lichuan**, Ningfei Zhu, and Shenglan Wu. Tropical cyclone's impacts on sea routes of china costal area during 1998-2007. *Journal of Central China Normal University (Natural Sciences)*, 43(04):27–30, 2009. (In Chinese with English Abstract).

## PARENT

1. Yuanqiao, Wen and Xin, Peng and **Lichuan, Wu** and Liang, Huang and Chunhui, Zhou. A method and system for selecting monitoring station sites for atmospheric pollutant emissions from port vessels. CN115032338B
2. Yuanqiao, Wen and Xin, Peng and **Lichuan, Wu** and Liang, Huang and Chunhui, Zhou and Changshi, Xiao. Buoy-based regional multi-ship marine fuel oil sulfur content monitoring device and method. CN113125682B

## SELECTED CONFERENCE PUBLICATIONS

1. ***Lichuan Wu***. Wave effects on coastal upwelling and water level. *Swedish Marine Modelling Symposium, December 1-2, 2022*
2. ***Lichuan Wu***, Victor George, Rutgersson Anna, Do we need to consider the wind input terms when parametering the wave decay by ice in the marginal ice zone? *SOOS-Swedish Southern Ocean Network Symposium, June, 2022*

3. ***Lichuan Wu*** and Qiao F. A turbulence closure scheme in the wave boundary layer and its application in a coupled model. *EGU, 2022*
4. ***Lichuan Wu*** and Qiao F. Wind profile in wave boundary layer and its application in a coupled atmosphere-wave model. *AGU, 2021*
5. Christian Dieterich, Matthias Groger, Julia Jeworrek, Kirsti Jylha, Erik Kjellstrom, H. E. Markus Meier, Taru Olsson, ***Lichuan Wu*** and Anna Rutgersson. Convective Snow Bands over the Baltic Sea in an Ensemble of Regional Climate Scenarios. *Baltic Earth Workshop, 2020*
6. Taru Olsson, Anna Luomaranta, Kirsti Jylha, ***Lichuan Wu***, Julia Jeworrek, Christian Dieterich, and Anna Rutgersson. Detecting sea-effect snowfalls on Finnish coastlines. *EMS Annual Meeting 2019*
7. ***Wu, L.***. Impact of ocean waves on the atmospheric boundary layer: wind stress and atmospheric mixing *International Workshop on Tropical Cyclone-Ocean interaction in the Northwest Pacific*, Jeju, Korea, June 2019 (Oral, invited)
8. ***Wu, L.***. A fully coupled atmosphere-wave-ocean model in the Baltic sea and North Sea *ACCME 3rd workshop on ocean surface waves and wave-coupled processes*, Hangzhou, China, Oct. 2018 (Oral)
9. ***Wu, L.***, Tihomir Hristov, Rutgersson, A. Vertical Profiles of WaveCoherent Momentum Flux and Velocity Variances in the Marine Atmospheric Boundary Layer. *23rd Symposium on Boundary Layers and Turbulence/21st Conference on Air-Sea Interaction*, Oklahoma City, USA, June,2018 (Oral)
10. ***Wu, L.***, Rutgersson, A., Sahlée, E. The impact of surface gravity waves on the atmospheric/oceanic numerical model simulations. *20th Air-Sea interaction conference*, Madison, USA, Aug.,2016 (Oral)
11. Jeworrek J., ***Wu, L.***, Rutgersson A. Numerical modelling of convective snow bands in the Baltic Sea area using atmosphere-ocean-wave coupled model systems. *Multiple drivers for Earth system changes in the Baltic Sea region*, Nida, Lithuania, Jun.,2016
12. ***Wu, L.***, Rutgersson, A., Sahlée, E. Modelling storm development and the impact when introducing waves, sea spray and heat fluxes *EGU General Assembly Conference*, Vienna, Austria. Apr.,2015 (Oral)
13. Rutgersson, A. ***Wu, L.*** Developing a swell-dependent surface roughness length for atmosphere-wave-ocean coupled models. *EGU General Assembly Conference*, Vienna, Austria. Apr.,2014
14. ***Wu, L.***, Rutgersson, A., Sproson, D., Sahlee, E. Comparison of air-sea momentum parameterizations in atmosphere-wave coupled system: Storm cases study. *3th International Lund Regional-Scale Climate Modelling Workshop* Lund, Sweden, Jun.,2014. (Oral)

## INVITED SEMINARS

- 2025.04: ECMWF, Bonn, Germany
- 2025.03: Reading University, UK
- 2025.01: Reading University, UK
- 2024.06: Polar Research Institute of China, China
- 2024.01: University of Miami, Miami, USA (online)
- 2023.01: Woods Hole Oceanographic Institution, USA (online)
- 2022.08: University of Cape Town, South Africa (online)
- 2019.02: University of Miami, Miami, USA
- 2018.05: Zhejiang University, Zhoushan, China
- 2018.03: Norwegian Meteorological Institute, Oslo, Norway
- 2018.10: Uppsala University, Uppsala, Sweden
- 2017.12, Sun Yat-Sen University, Zhuhai, China
- 2017.08, Norwegian Meteorological Institute, Bergen, Norway
- 2017.06, The First Institute of Oceanography (FIO), Ministry of Natural Resources, China
- 2017.05, Swedish Meteorological and Hydrological Institute, Norrkoping, Sweden
- 2017.05, Hohai University, Nanjing, China
- 2017.05, Nanjing University of Information Science & Technology Nanjing, China

## REFEREE ASSIGNMENTS

- Board member of the CIM and AI4TekNat at Uppsala University
- Reviewer of the research grant proposals: Research Grants Council (Hong Kong); and Natural Environment Research Council (NERC), UK; National Research and Development Agency, Chile; Research Ireland Centre for Applied Geosciences (Ireland)
- PhD defence examiner of 5 PhD students
- Guest Editor of Ocean Modelling, Frontiers In Marine Science, and Journal of Marine Science and Engineering

- Editorial Board Member of Ocean-Land-Atmosphere Research (2023-), Journal of Atmospheric Science Research (2019-2020), Atmosphere (2021-)
- Reviewer of Nature Communication, Journal of Climate, Scientific Report, Geophysical Research Letters, Journal of the Atmospheric Sciences, Boundary-Layer Meteorology, Ocean Modelling, JGR-Oceans, JGR-Atmosphere, Journal of Advances in Modeling Earth Systems, Earth and Space Science, Ocean Engineering, Ocean dynamics, Ocean science, The Cryosphere, Tellus-A, Q J R Meteorol Soc., Journal of Operational Oceanography, Journal of Oceanology and Limnology, Progress in Oceanography, Atmosphere, Remote Sensing, International Journal of Remote Sensing, Journal of Marine Science and Engineering, Geophysica, and Frontiers Marine Science.

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